

Lecture 4.1: Constructing Regular Expressions - Part 1

Presenter: Azwad Anjum Islam (AAI)

Scribe: Mujtahid Al-Islam Akon (AKO)

In this lecture, you will learn to construct regular expressions through some examples.

Problem 1

Let, L be a language, where L is the set of **all binary strings**. Is L a regular language? If yes, what is its regular expression?

Answer

A valid binary string means **one or more occurrences** of 0 or 1. Recall that one or more occurrences means positive closure. So, the result should be $(0|1)^+$.

As we have found a regular expression, the language must be regular.

Problem 2

Let, $L = \{a, aa, aaa, aaaa, aaaaa, \dots, ab, abab, ababab, abababab, abababab, \dots\}$

Is L a regular language? If yes, what is its regular expression?

Answer

$L = \{a, aa, aaa, aaaa, aaaaa, \dots, ab, abab, ababab, abababab, abababab, \dots\}$

Notice that the strings of this language consists of either **one or more occurrences of a** or **one or more occurrences of ab** .

one or more occurrences of $a = a^+$

one or more occurrences of $ab = (ab)^+$

ORing them, we get the result, **$a^+|(ab)^+$**

As we have found a regular expression, the language must be regular.

For the following examples, assume the alphabet set, $\Sigma = \{0, 1\}$.

Problem 3

What if L was the language of all binary strings that **starts with 0**?

Answer

To start a binary number with 0, we need a zero first. This zero should be followed by any binary string of length ranging from zero to anything. Binary string with zero or any length means $(0|1)^*$. After prepending a 0, we get the result, **$0(0|1)^*$**

Problem 4

What about the language of all binary strings that ends with two consecutive **1s**?

Answer

A RegEx for any binary string that can be generated including empty string (why?) is $(0|1)^*$. Appending two 1s we get the desired result, **$(0|1)^*11$** .

To be continued ...
